Intro to Git & GitHub

Week 1 - 06/15/2026

LEARNING GOALS:

Understand the “Reproducibility Crisis”
Learn the difference between Git & GitHub
Familiarize yourself with essential vocabulary

How do you know if work is trustworthy?

A Case Study: *Jonathan Pruitt*

University investigation found prominent spider biologist fabricated, falsified data

Co-authors of now-retracted papers from Jonathan Pruitt were sent summarized findings that the scientist violated McMaster University's research integrity policy

11 MAY 2023 · 7:20 PM ET · BY CLAUDIA LÓPEZ LLOREDA

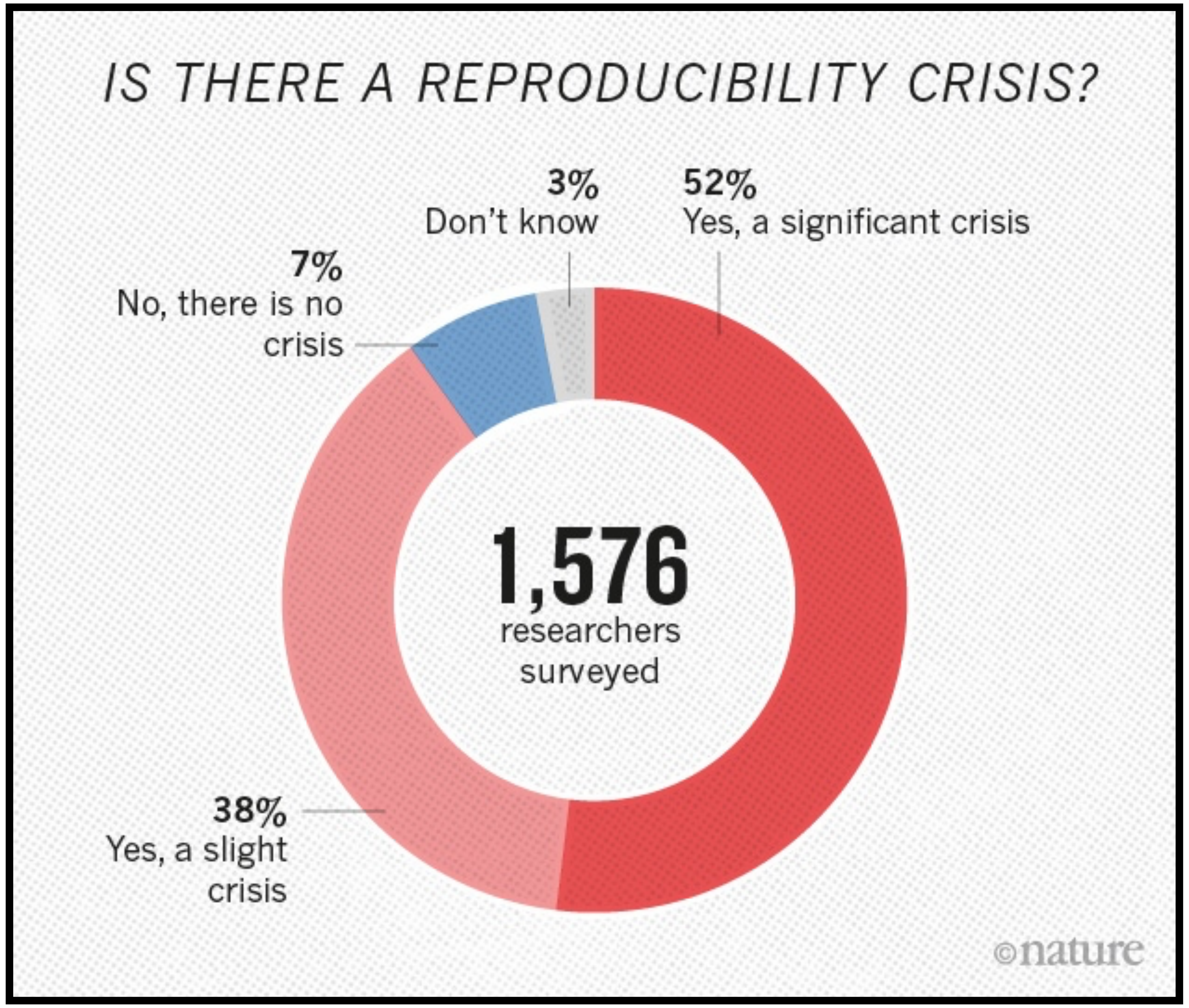
Behavioral ecologist Jonathan Pruitt's PhD dissertation withdrawn

Fraud is **hard to catch** since scientists **tend to trust** one another

But *really*, how common are mistakes?

The Reproducibility Crisis

A phenomena in scientific fields where *published results cannot be reliably replicated*. Most associated with medical research, but shows up across many fields.



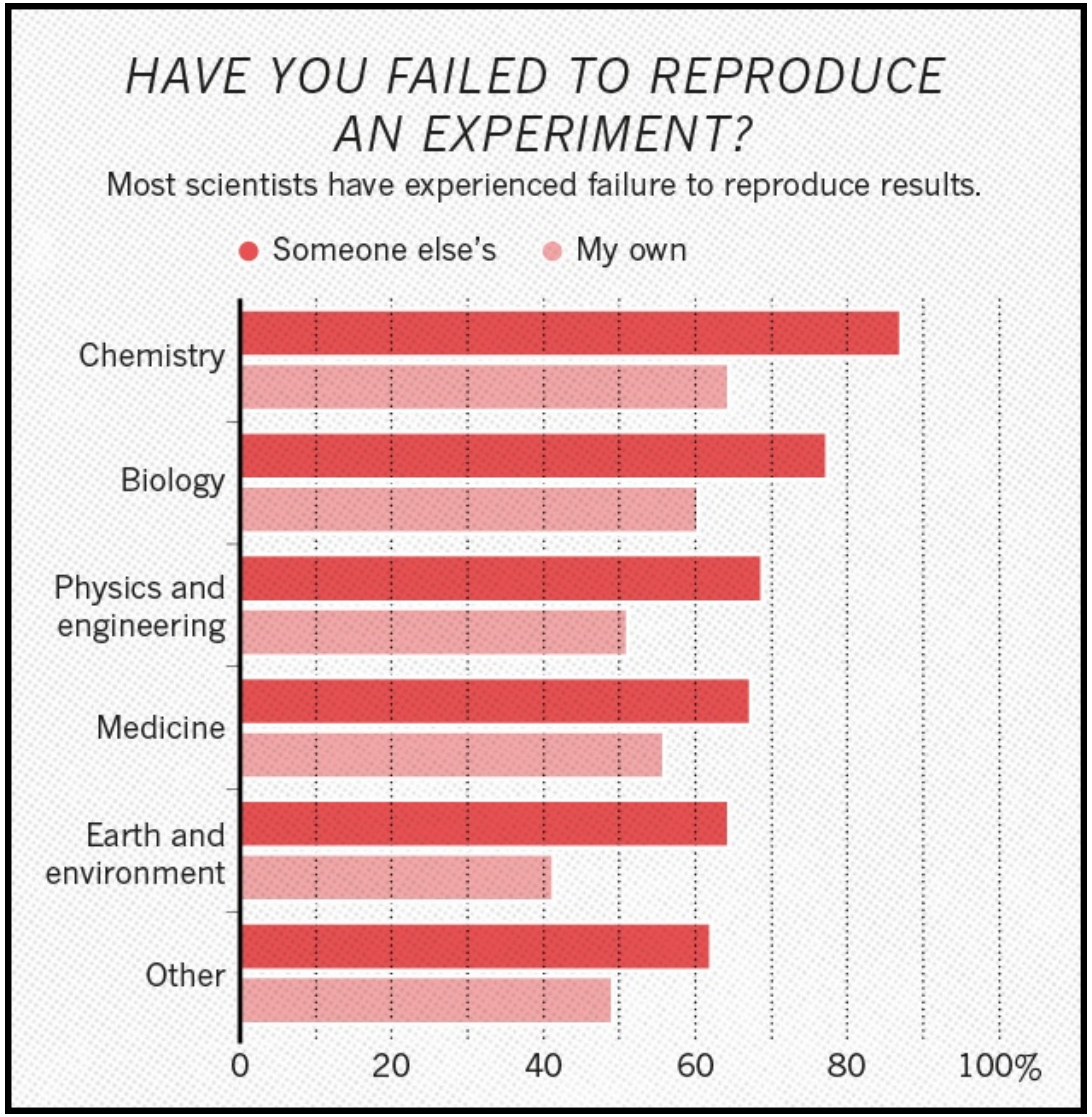
from Monya Baker; "1,500 scientists lift the lid on reproducibility"

The Reproducibility Crisis

A phenomena in scientific fields where *published results cannot be reliably replicated*. Most associated with medical research, but shows up across many fields.

Across many disciplines, scientists sometimes ***fail*** to ***replicate results***.

This includes **their own experiments**.



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How do I establish trust?

How Do You Write Robust, Reproducible Code?

Ensuring all code follows a “style guide”

Unit / Functional Testing

Providing comments + docstrings

Explicitly handling errors / exceptions

Writing small single-purpose functions

⋮

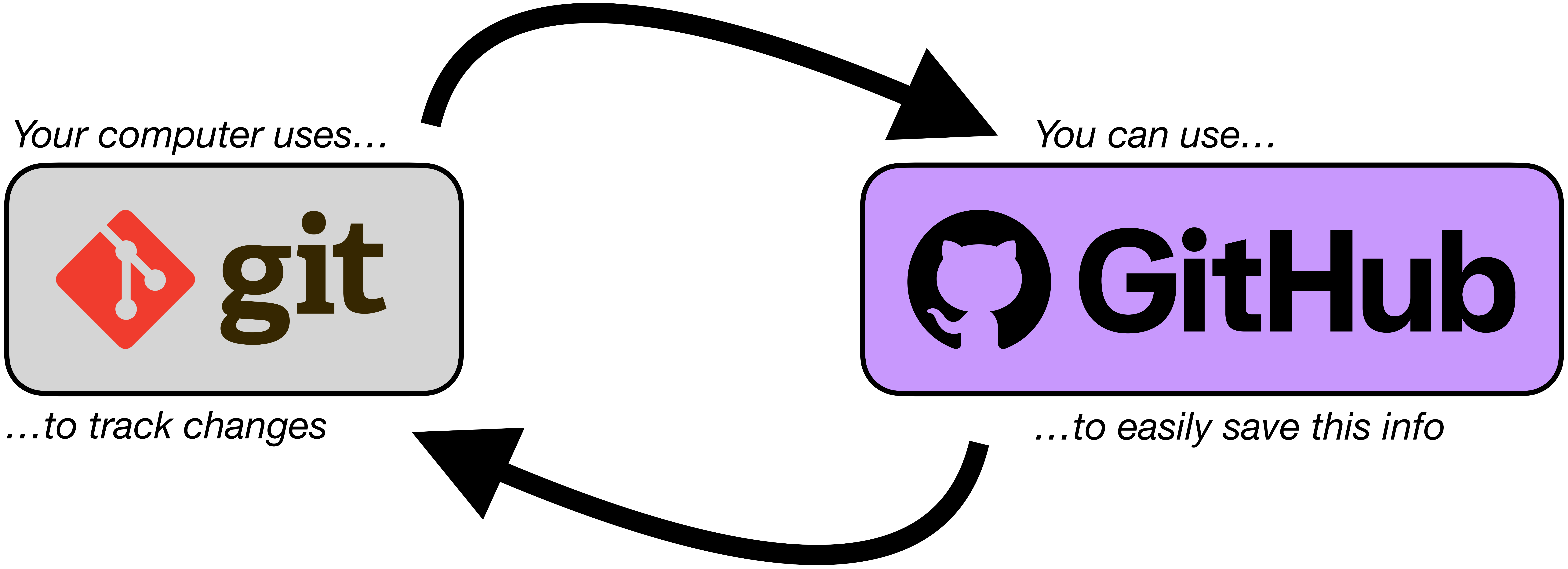
Each of these topics will be discussed this Summer!

A Very Basic Overview of Git / GitHub

git *is not the same as* GitHub

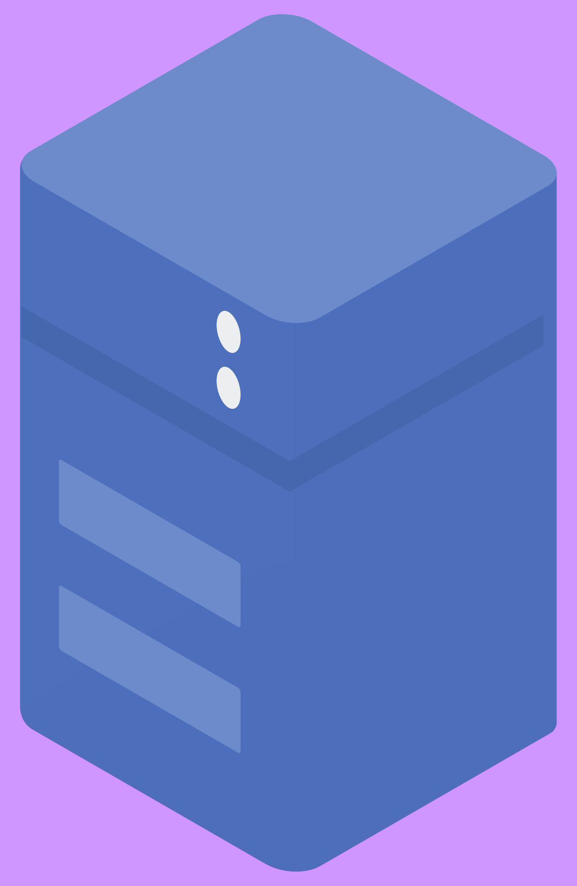
- version tracking software
- runs entirely on your device
- open-source project

- file hosting service
- runs entirely on remote servers
- owned by Microsoft


Your computer uses…
git
…to track changes
You can use…
GitHub
…to easily save this info

Two Types of Repositories

Local Repository



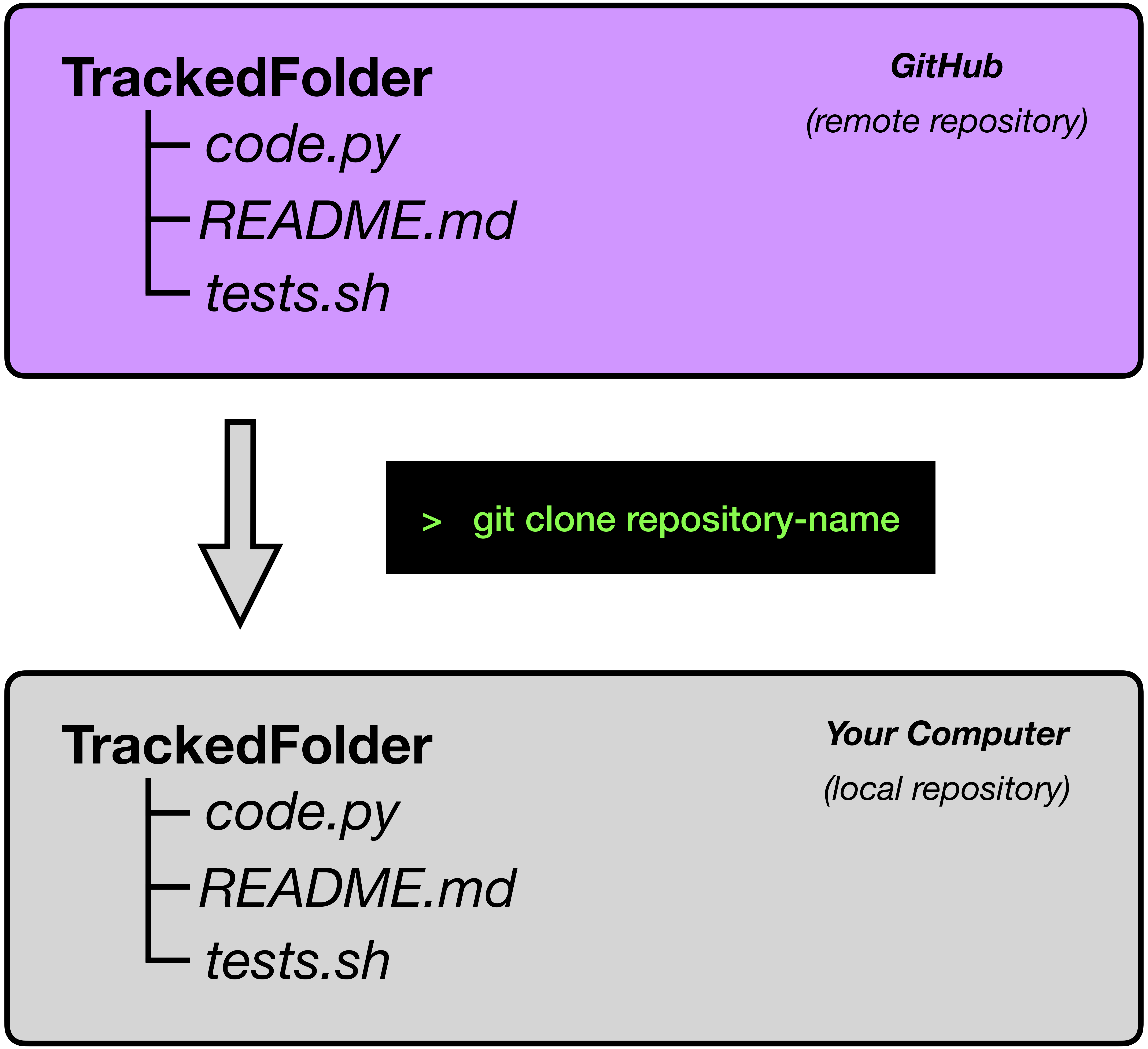
Remote Repository

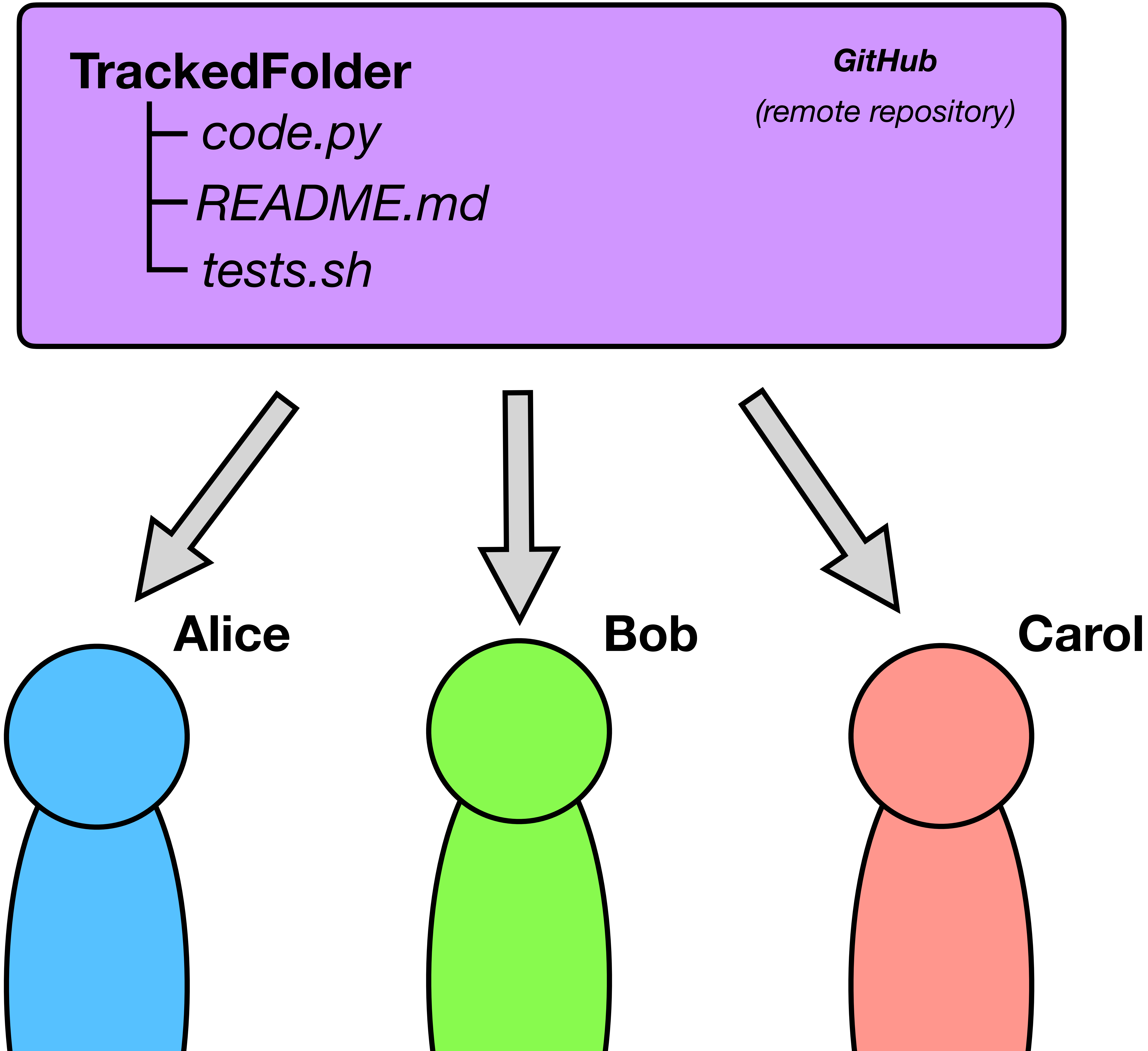
Setting up a local repository

If a repository already exists, then you need to ***clone*** the ***remote repository***.

This command will create a tracked folder in your terminal's current directory.

Once cloned, you should not move the folder! This will break Git's ability to track files.

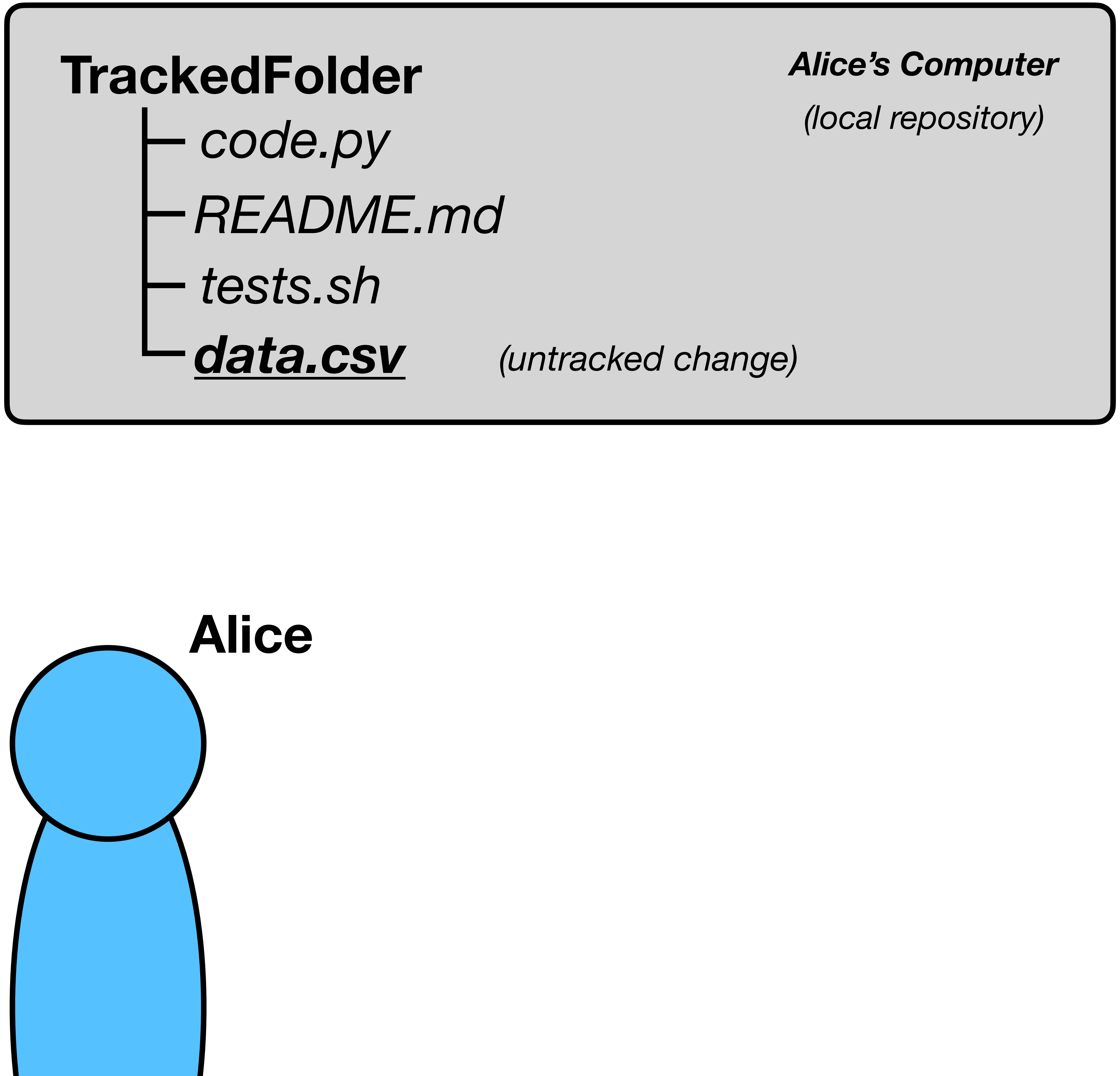




Can multiple people clone?

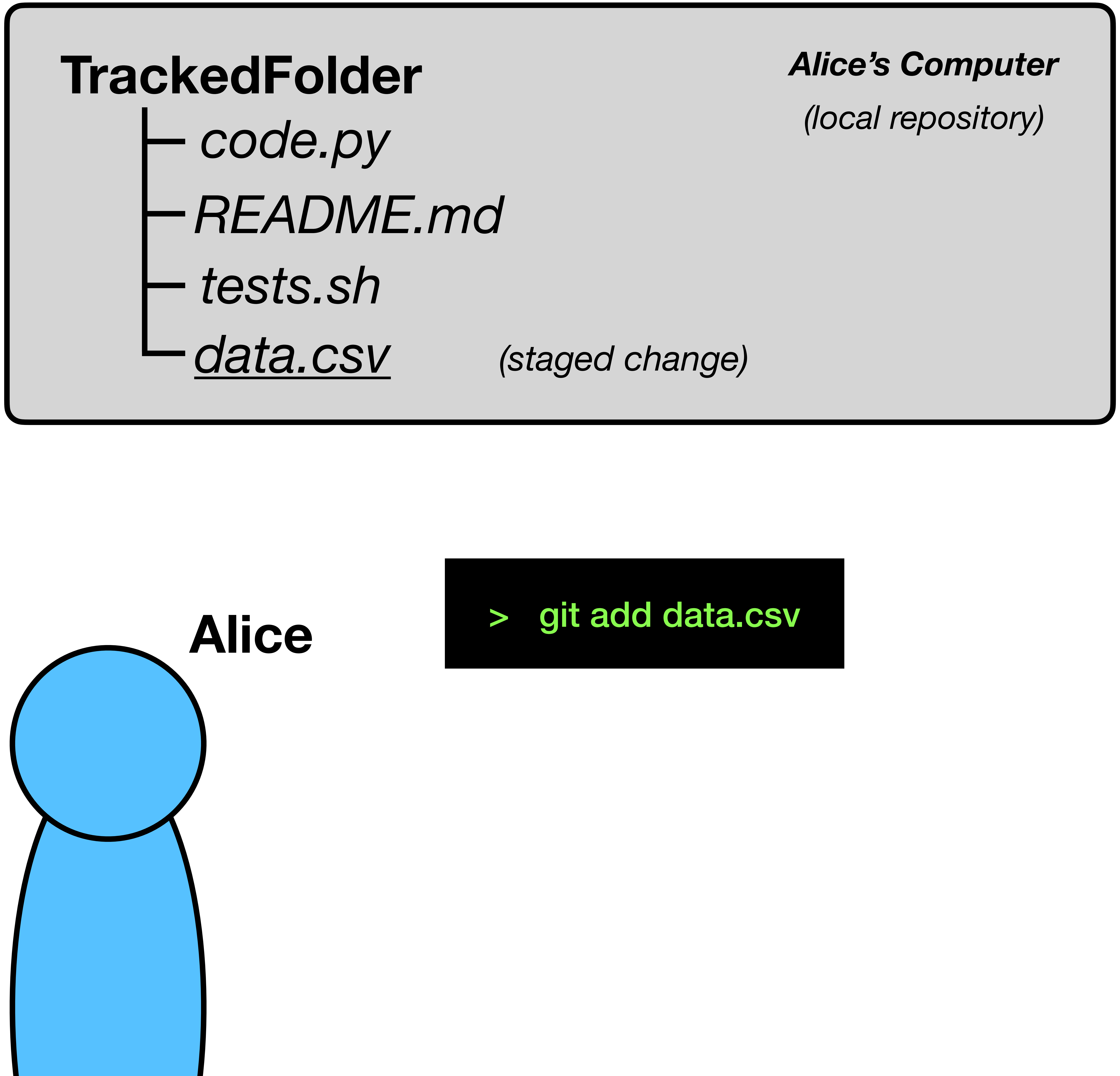
We will create three imaginary people to help explain how Git & GitHub work.

Since Alice, Bob, and Carol all cloned the remote repository, they all have the same files on their own devices.



How do I make changes?

Let's say Alice added a file called "data.csv" to her local repository. At this point, Git has no idea what you have done!



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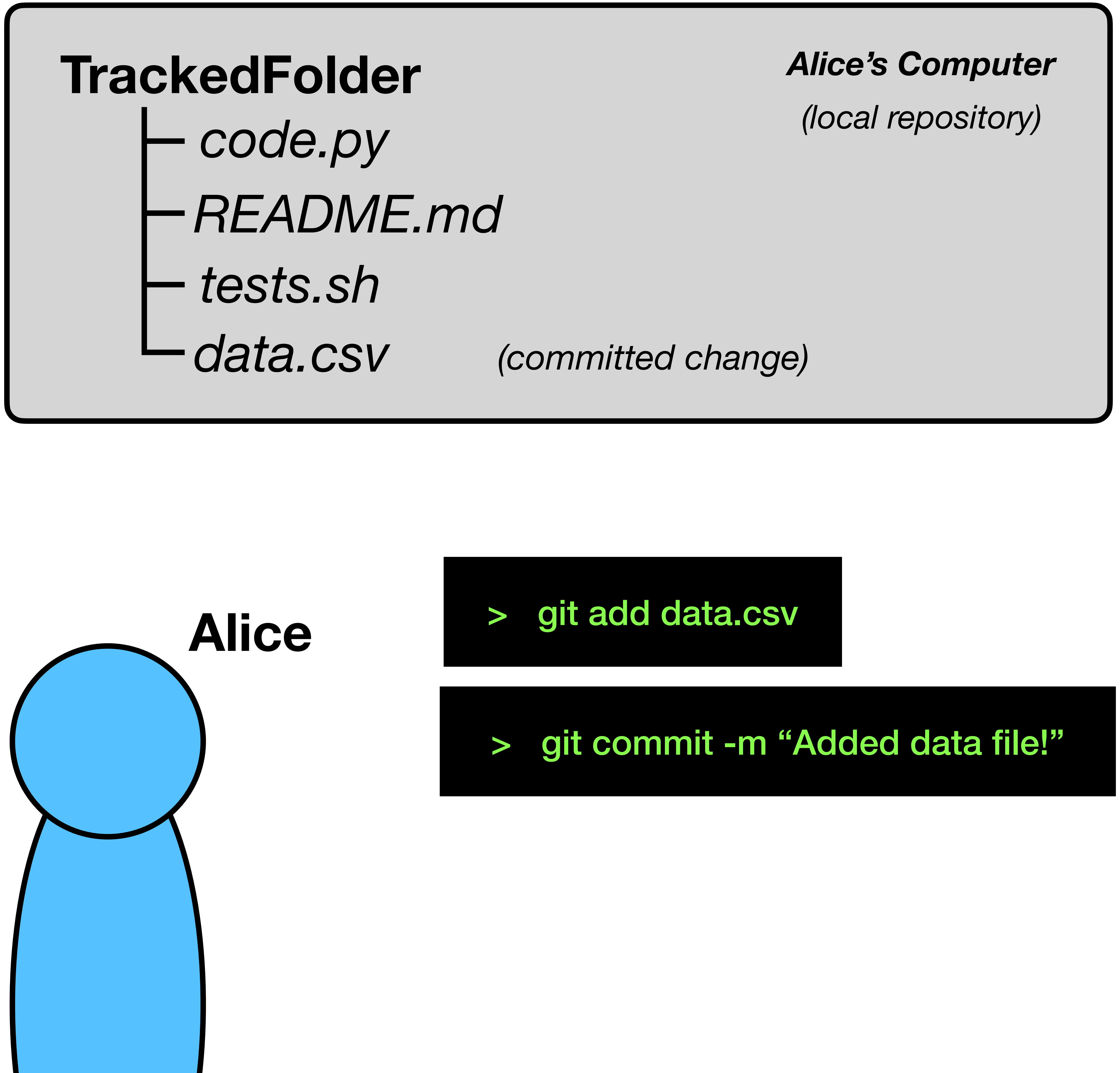
To tell Git that you've made a change, you need to ***add*** the file. This prepares Git for making a "save state," but has not done so yet.



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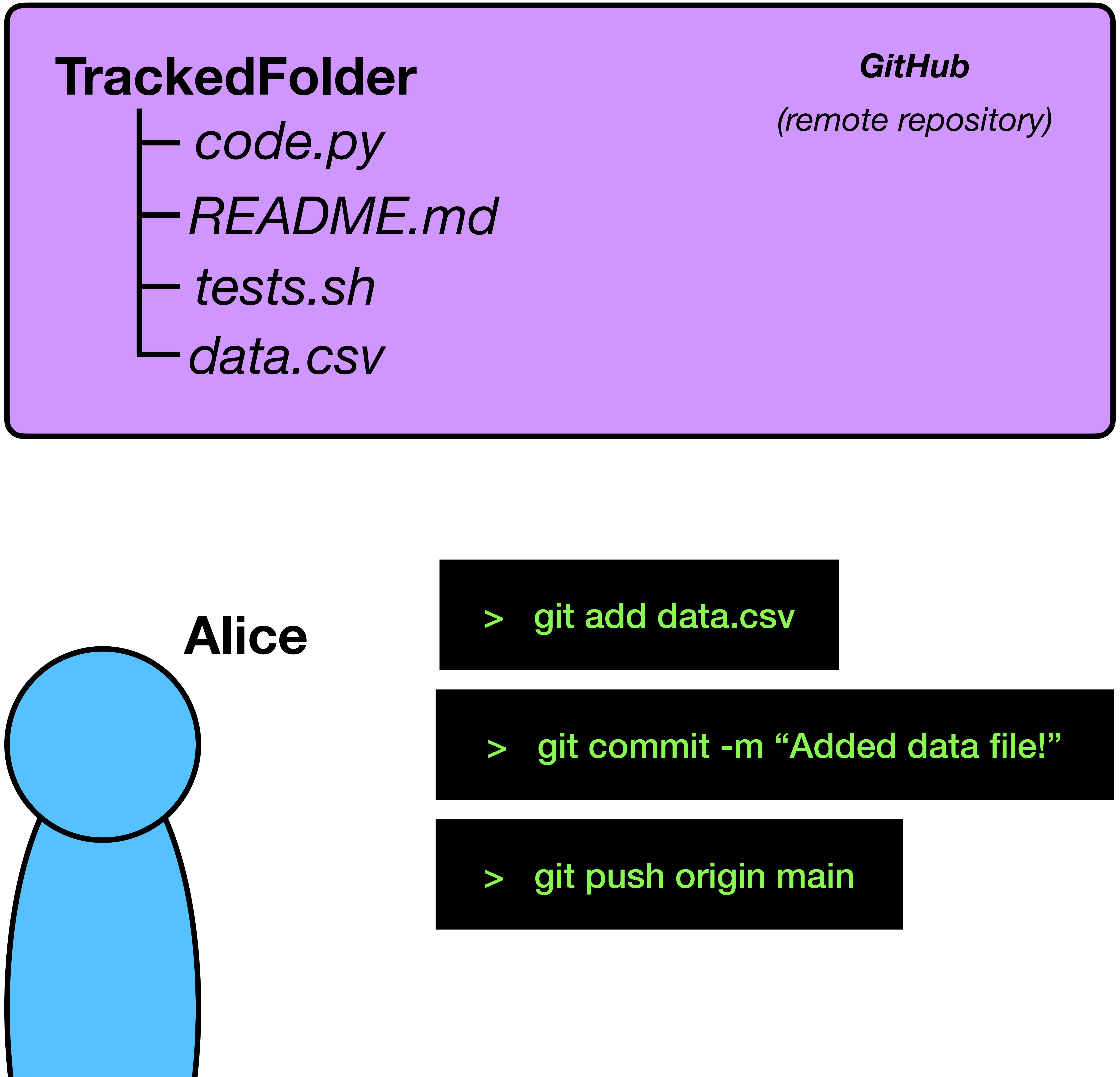
After adding all changed files, you need to ***commit*** it to finalize the change.



Updating the remote repo.

Once you finished all your changes, you can update the ***remote repository*** by ***push***ing your local version.

This can take a couple seconds to show up on GitHub, but is usually very fast.

TrackedFolder

├ *code.py*
├ *README.md*
├ *tests.sh*
└ *data.csv*

GitHub

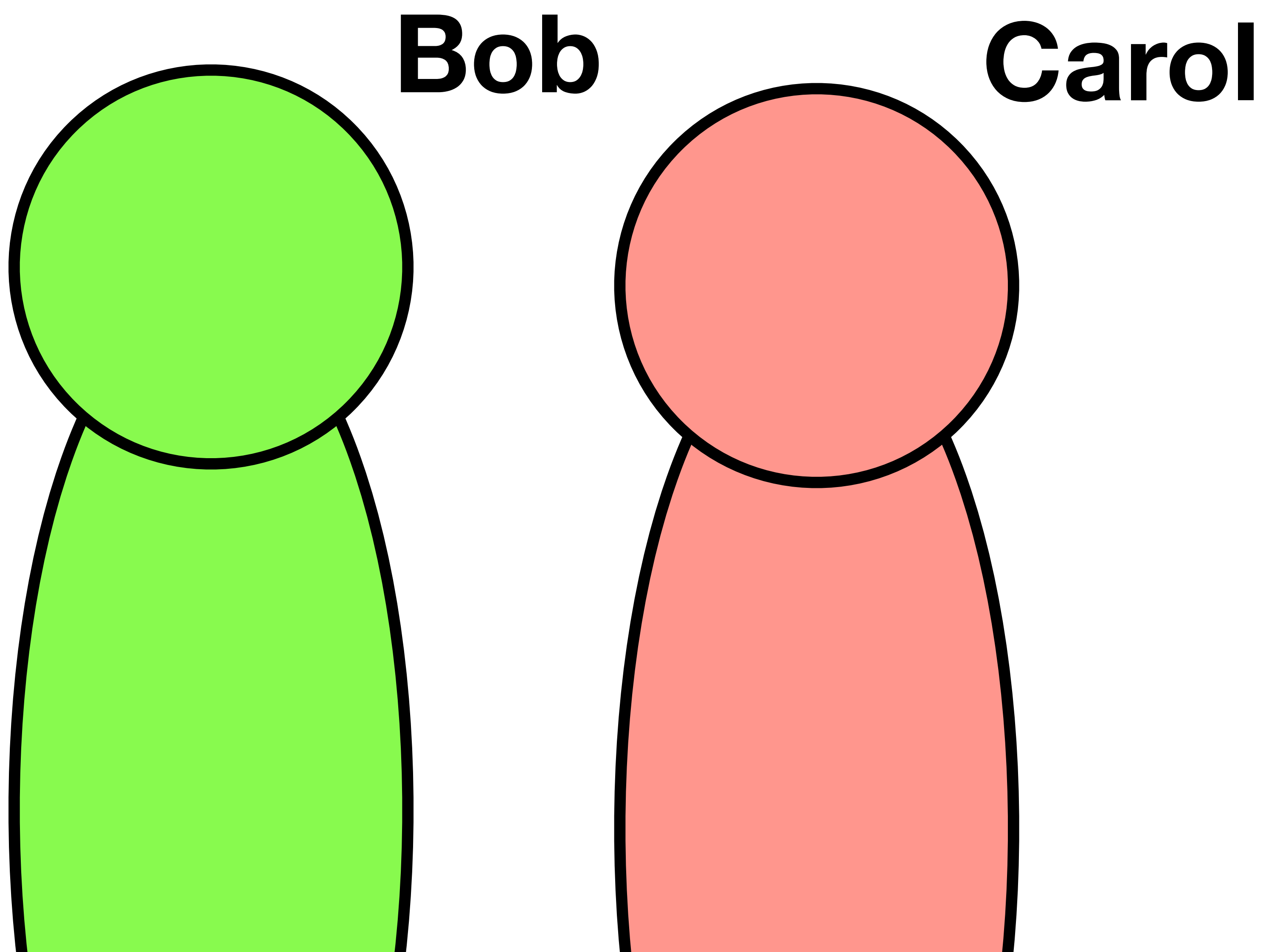
(remote repository)

TrackedFolder

├ *code.py*
├ *README.md*
└ *tests.sh*

Bob / Carol Computer

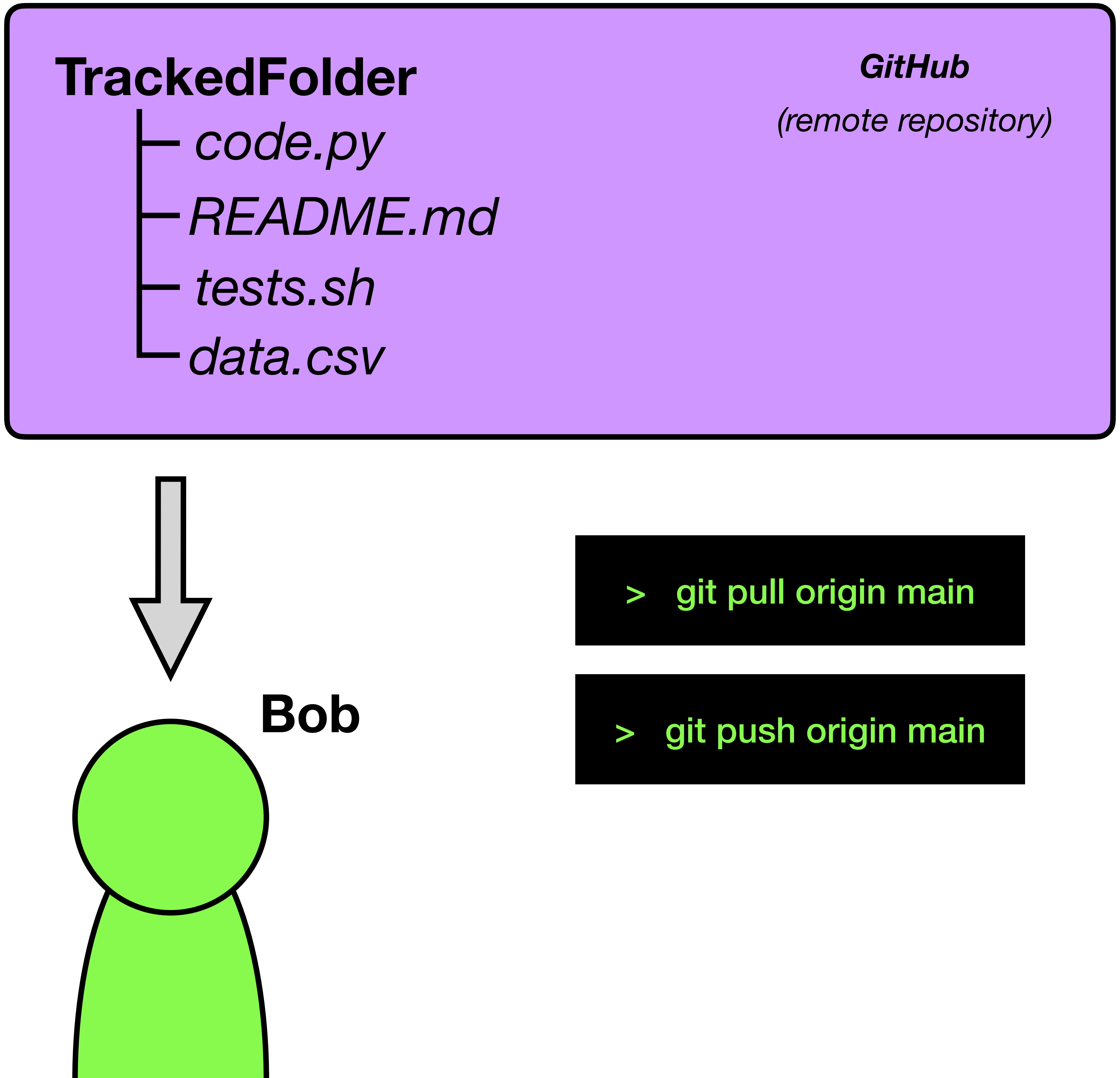
(local repository)



A discrepancy in repositories...

Remember, Bob and Carol only have the version of the code that existed before Alice made her changes!

If they were to try and ***push*** right now, they would get a bunch of errors and warnings.




```
> git pull origin main
```

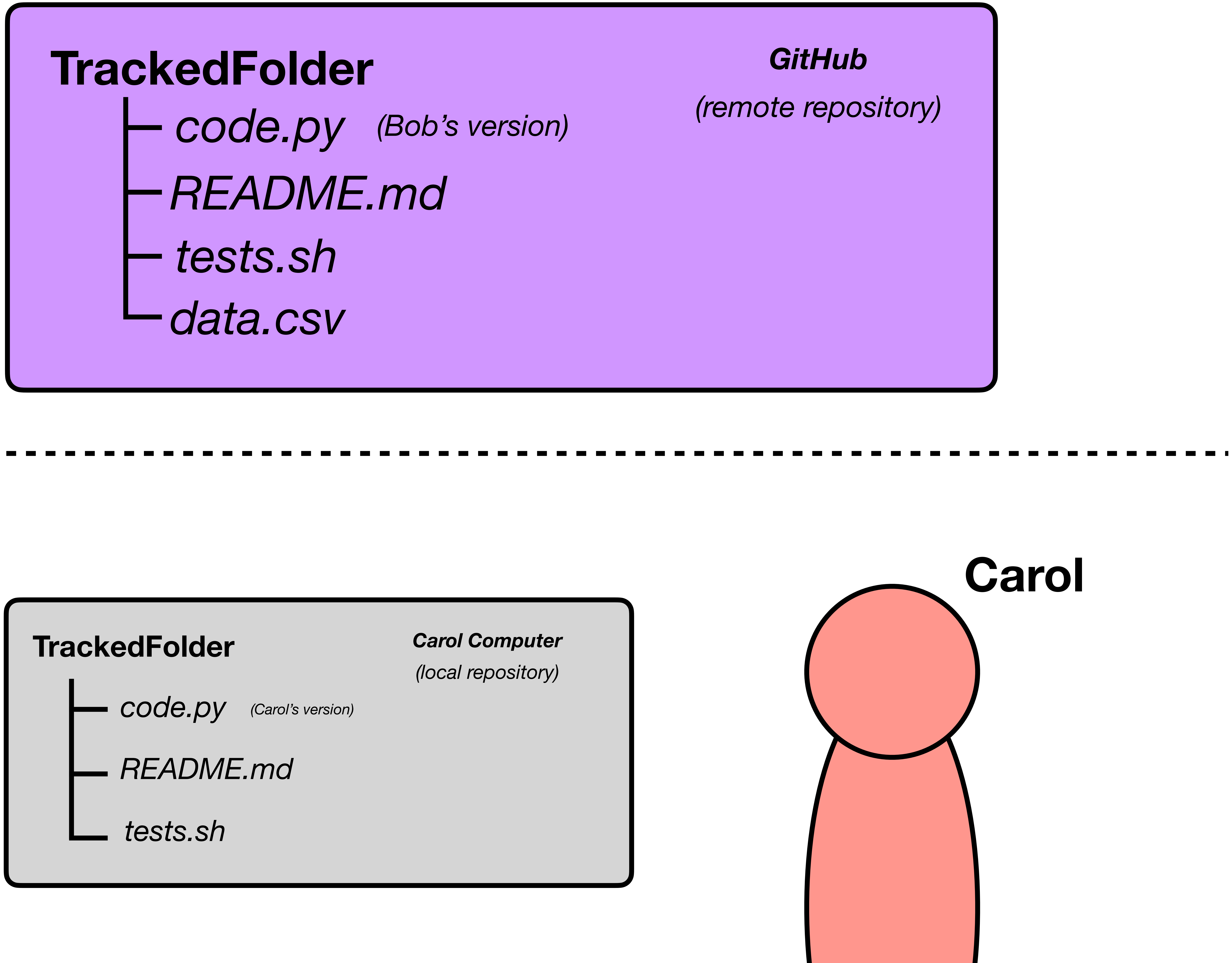
```
> git push origin main
```

Updating your local repository

Remember, Bob and Carol only have the version of the code that existed before Alice made her changes!

If they were to try and ***push*** right now, they would get a bunch of errors and warnings.

For relatively simple conflicts, you can fix this discrepancy by ***pull***ing the remote repository.



Trying to Handle Conflicts

Let's say Carol and Bob both made changes to overlapping sections of *code.py*. It isn't easy to know which changes to keep…

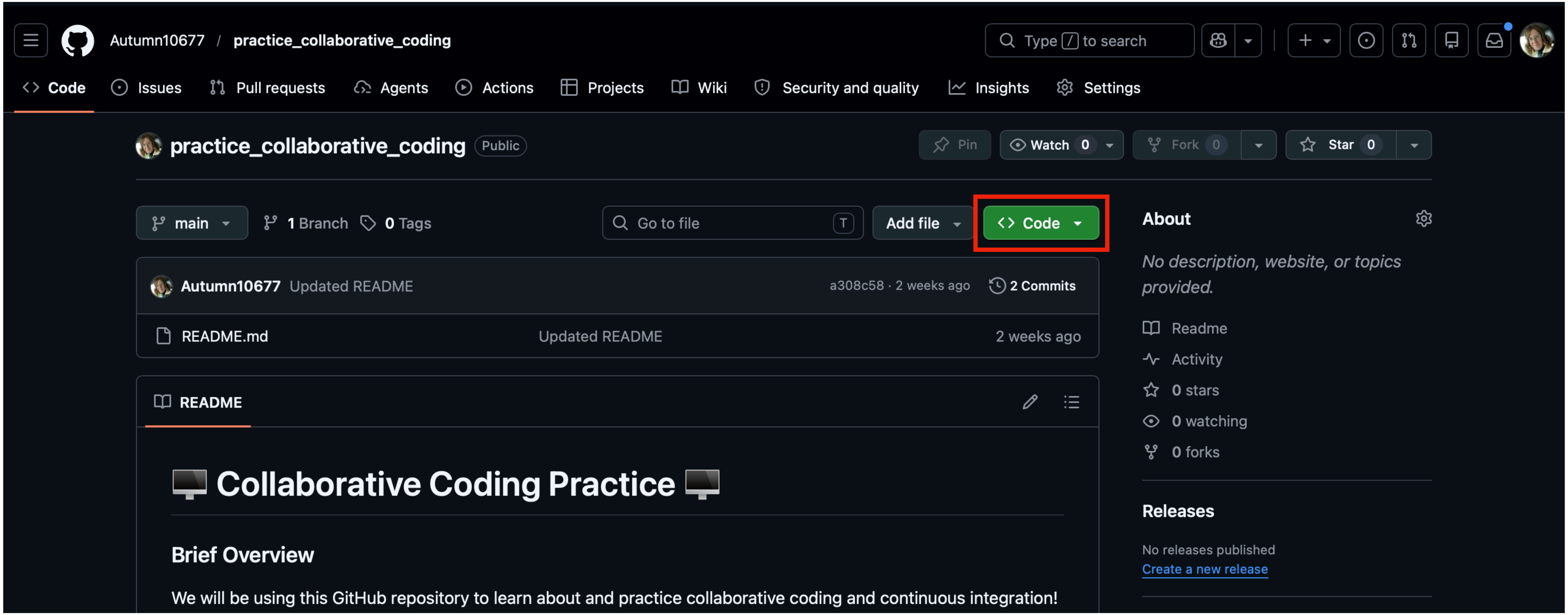
If you use VS Code, you will be asked to select which changes to keep for every detected conflict.

In the future, we will talk about how using GitHub can prevent thorny conflicts from arising!

Continuous Integration Rule:

Merge your changes **frequently** so that file conflicts are small

Getting Started: *Let's Clone a Repository!*



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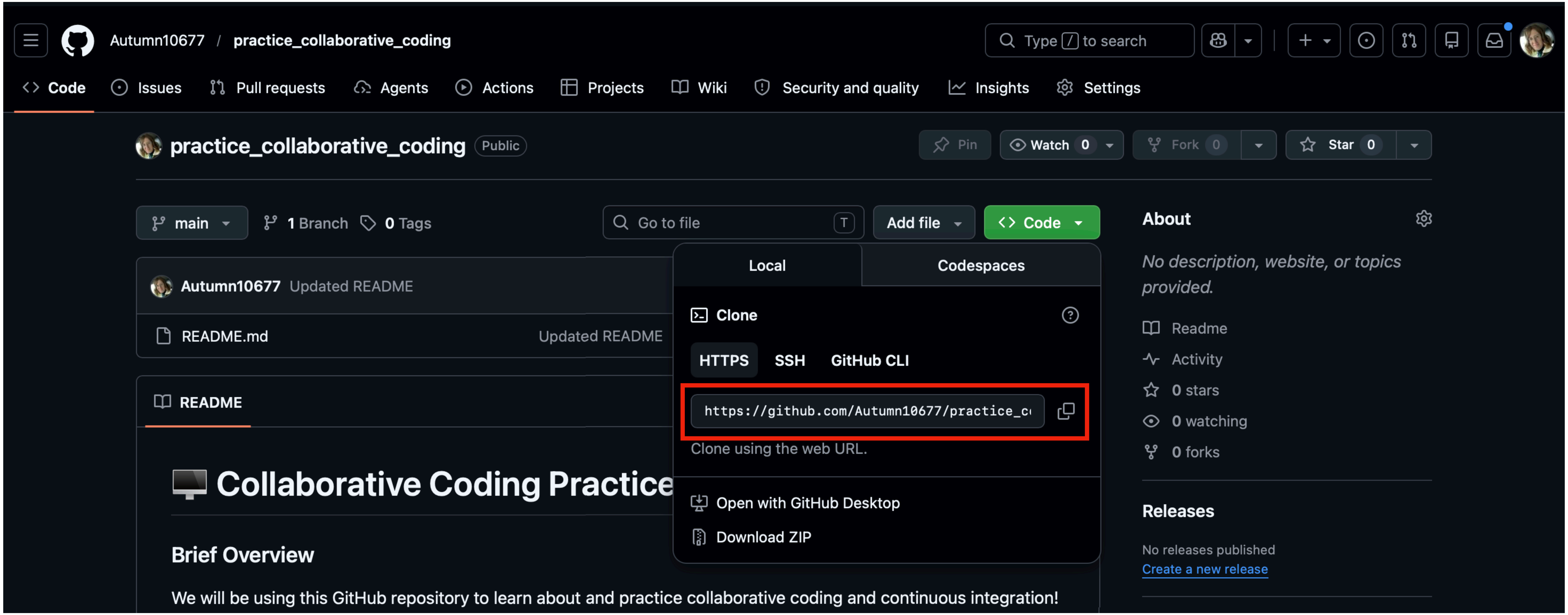
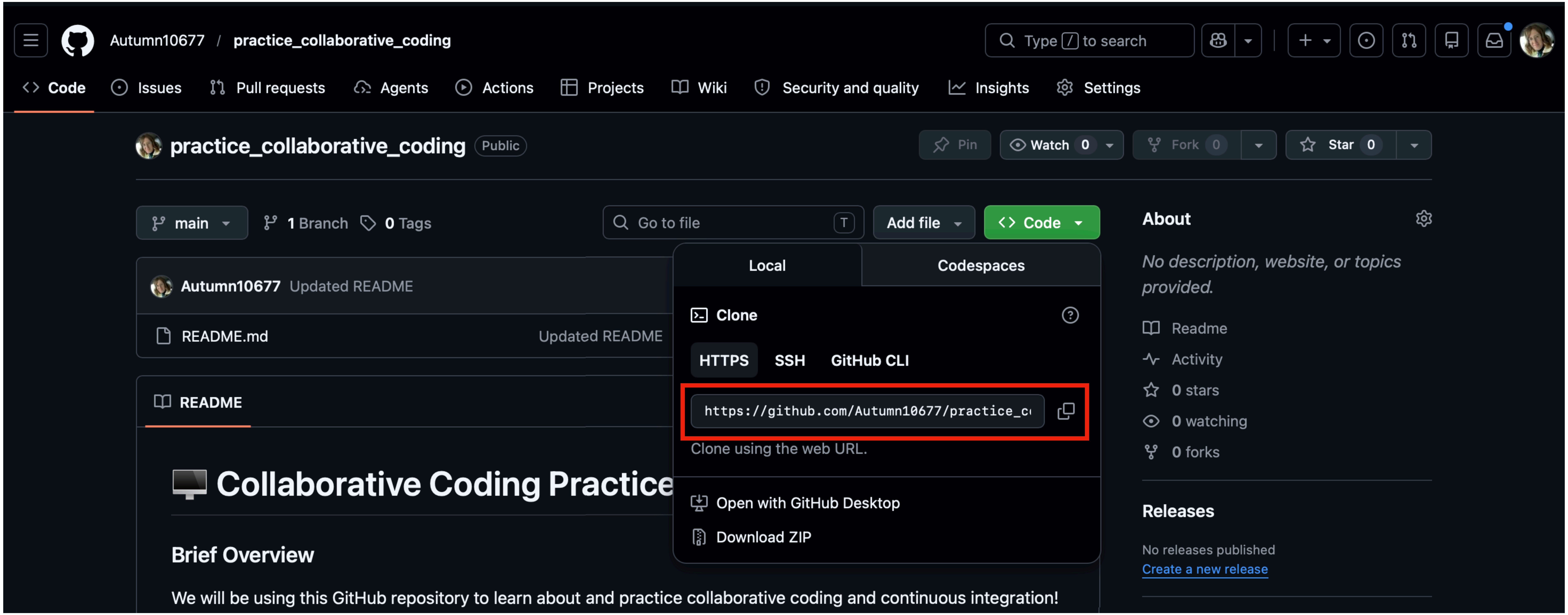
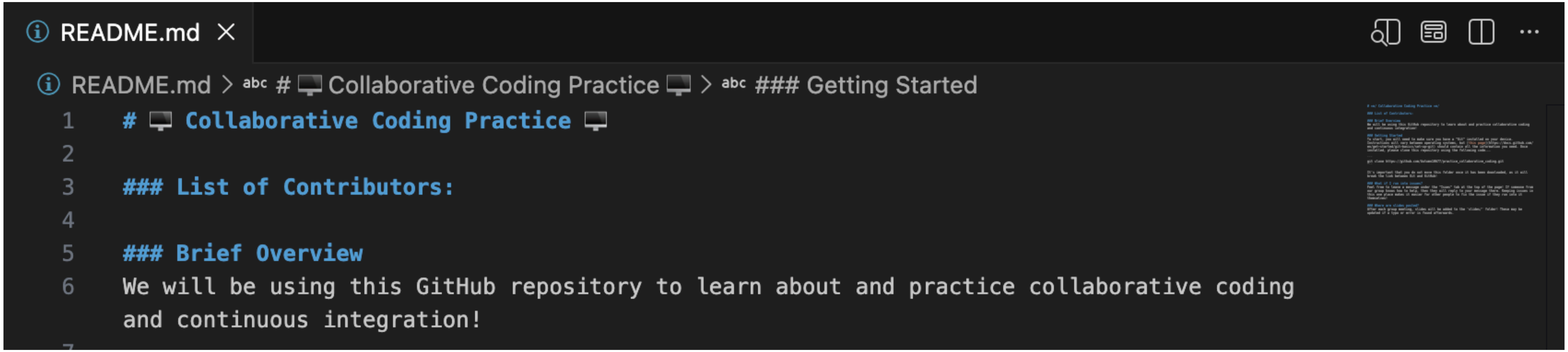


Getting Started: *Let's Clone a Repository!*

```
>  git clone https://github.com/Autumn10677/practice_collaborative_coding.git
```



Getting Started: *Updating the README.md*



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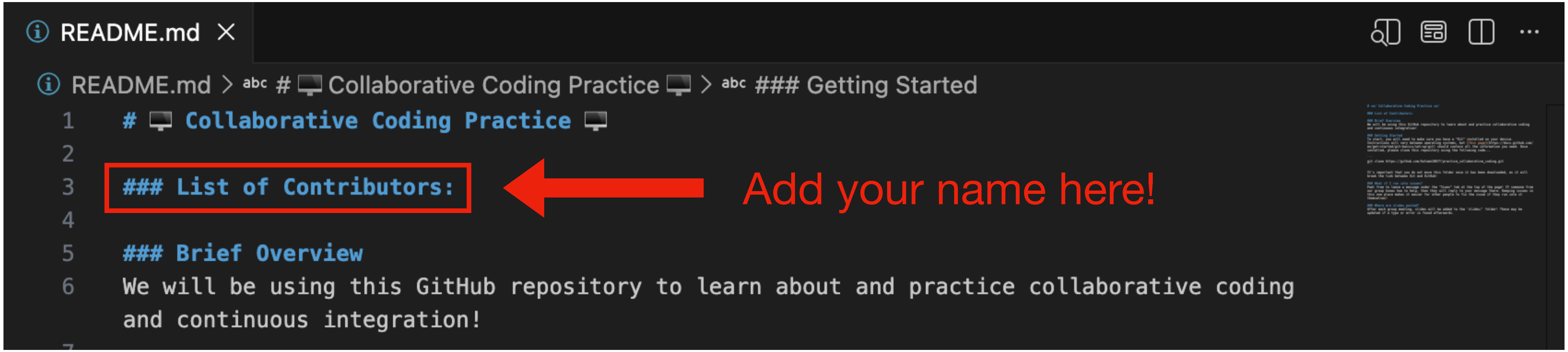

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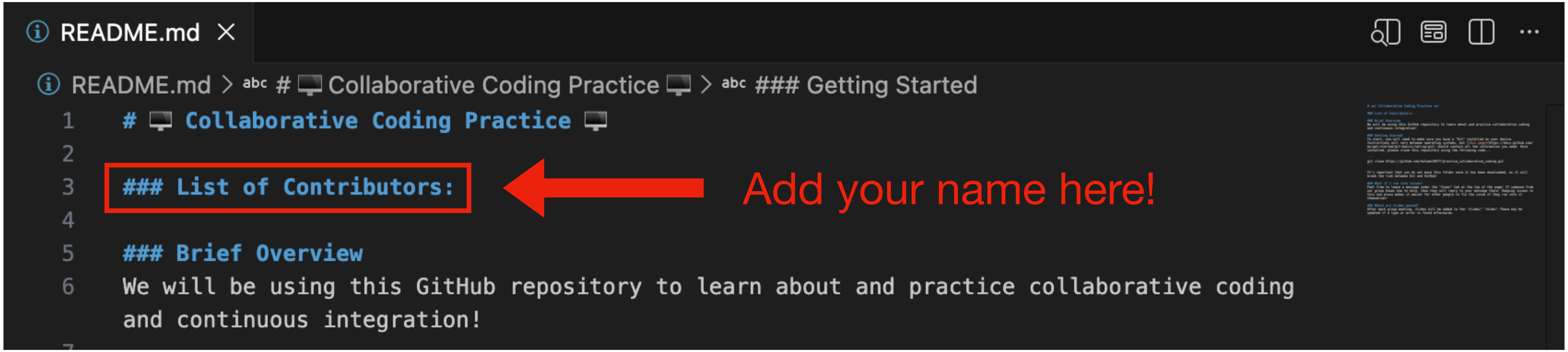


```
> git add README.md
```

```
> git commit -m “Added ‘___’ to README”
```

```
> git push origin main
```

Save and **upload** your changes!

A Handy Cheat Sheet

`git…`

`…clone`	First-time **download and setup** of local a repository
`…add`	**Stages** a changed file for saving
`…commit`	Create a **new save state** with all new changes
`…push`	**Update remote** repository **using local** changes
`…pull`	**Update local** repository **using remote** changes

Next Time…

Introduction to Branches & Style Guides